MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. If complex numbers $Z_1, Z_2, \ldots Z_n$ satisfy the equation $4Z^2 + \bar{Z} = 0$, then $\sum_{i=1}^{n} |Z_i|^2$ is equal to

(1) $\frac{3}{16}$ (2) $\frac{3}{64}$

(3) $\frac{9}{64}$ (4) $\frac{1}{16}$

Answer (1)

Sol. $4z^2 + \bar{z} = 0$

If $z = 0$ then it is a solution

$\Rightarrow |z| = 0$

Now, $z \neq 0$

Multiplying by z we get

$4z^3 + |z|^2 = 0$

$\Rightarrow 4z^3 = -|z|^2$

Taking modulus both sides

$4|z|^3 = |z|^2 \Rightarrow |z|^2(4|z| - 1) = 0$

$\Rightarrow |z| = \frac{1}{4}$

$\Rightarrow 4z^3 = \frac{-1}{16} \Rightarrow z^3 = \frac{-1}{64}$

$\Rightarrow z = \frac{-1}{4}, \frac{-1}{4}w, \frac{-1}{4}w^2$

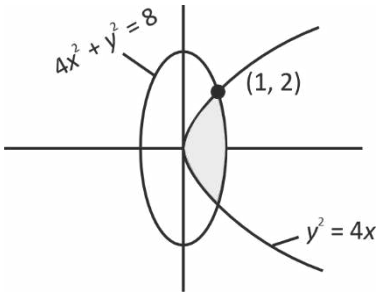
2. Area enclosed by $4x^2 + y^2 \leq 8$ and $y^2 \leq 4x$ is (in square unit) is

(1) $\left(\pi + \frac{4}{3}\right)$ sq. unit (2) $\left(\pi - \frac{4}{3}\right)$ sq. unit

(3) $\left(\pi + \frac{2}{3}\right)$ sq. unit (4) $\left(\pi - \frac{2}{3}\right)$ sq. unit

Answer (3)

Sol.



$$\text{Area} = 2\int_0^2 \sqrt{\frac{8-y^2}{4}} - \frac{y^2}{4}\,dx$$

$$= \int_0^2 \left(\sqrt{8-y^2} - \frac{y^2}{2}\right)dy$$

$$= 4\sin^{-1}\left(\frac{y}{2^{3/2}}\right) + \frac{y\sqrt{8-y^2}}{2} - \frac{y^3}{6}\Bigg]_0^1$$

$$= 4\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + \frac{2\times 2}{2} - \frac{8}{6}$$

$$= \frac{4\pi}{4} + 2 - \frac{4}{3}$$

$$= \left(\pi + \frac{2}{3}\right) \text{sq. unit}$$

3. Mean deviation about median for k, $2k$, $3k$,……, $1000k$ is 500, then the value of k^2 is

(1) 4

(2) 9

(3) 16

(4) 1

Answer (1)

Sol. data set

k, $2k$, $3k$, ..., $1000k$.

$$\text{Median} = \frac{500k + 501k}{2} = 500.5k$$

Mean-deviation about median



$$500=\frac{|500.5k-k|+|500.5k-2k|+...+|1000k-500.5k|}{1000}$$

$$=\frac{2\left[499.5k+498.5k+...+0.5k\right]}{1000}$$

$$250000=\frac{500}{2}\left[0.5+499.5\right]k$$

$$\Rightarrow \boxed{k=2}$$

$$\boxed{k^2=4}$$

4. If $4x^2 + y^2 < 52$, $x, y \in I$ then number of ordered pairs (x, y) is

(1) 67 (2) 87
(3) 77 (4) 38

Answer (3)

Sol. $4x^2 + y^2 < 52$, $x, y \in I$

$\Rightarrow 4x^2 < 52 \Rightarrow x^2 < 13 \Rightarrow x^2 \in \{0, 1, 4, 9\}$

If $x^2 = 0 \Rightarrow y^2 < 52 \Rightarrow y^2 \in \{0, 1, 4, 9, 16, 25, 36, 49\}$

$1 \times (1 + 2 \times 7) = 15$

If $x^2 = 1 \Rightarrow y^2 < 48 \Rightarrow y^2 \in \{0, 1, 4, 9, 16, 25, 36\}$

$2 \times (1 + 2 \times 6) = 26$

If $x^2 = 4 \Rightarrow y^2 < 36 \Rightarrow y^2 \in \{0, 1, 4, 9, 16, 25\}$

$2 \times (1 + 2 \times 5) = 22$

If $x^2 = 9 \Rightarrow y^2 < 16 \Rightarrow y^2 \in \{0, 1, 4, 9\}$

$2 \times (1 + 2 \times 3) = 14$ $\Rightarrow$ 77 pairs

5. $x - ny + z = 6$

$x - (n - 2)y + (n + 1)z = 8$

$(n - 1)y + z = 1$

Let n = number on the dies when rolled randomly then P (that system equation has unique solution) $= \frac{k}{6}$, then sum of value of k and all possible value of n is

(1) 22 (2) 21
(3) 20 (4) 24

Answer (4)

Sol. For system of equation to have unique solution

$\Delta \neq 0$

$$\begin{vmatrix}1 & -n & 1\\ 1 & 2-n & n+1\\ 0 & n-1 & 1\end{vmatrix} \neq 0$$

$\Rightarrow n^2 - n - 2 \neq 0$

$(n - 2)(n + 1) \neq 0$

$\Rightarrow n \neq 2$

$\therefore$ possible values of n is {1, 3, 4, 5, 6}

P(unique solution) = $\frac{5}{6} = \frac{k}{6}$

$\Rightarrow k = 5$

$\therefore$ sum of all possible value of n and k is

$1 + 3 + 4 + 5 + 6 + 5 = 24$

6. If $\lim\limits_{x\to 0}\frac{e^{(a-1)x}+2\cos(bx)+e^{-x}(c-1)}{x\cos x-\ln(1+x)}=2$

Then the value of $a^2 + b^2 + c^2$ is

(1) 8 (2) 9
(3) 10 (4) 11

Answer (3)

Sol. $$\lim_{x\to 0}\left(1+x(a-1)+\frac{(x(a-1))^2}{2!}+...\right)+2\left(1-\frac{b^2x^2}{2!}+\frac{b^4x^4}{4!}\right)$$
$$+\frac{\left(1-x+\frac{x^2}{2!}-\frac{x^3}{3!}+...\right)(c-1)}{x\left(1-\frac{x^2}{2!}+\frac{x^4}{4!}+...\right)-\left(x-\frac{x^2}{2}+\frac{x^3}{3}+...\right)}$$

$$\Rightarrow \frac{(1+2+(c-1))+x(a-1-c+1)+x^2\left(\frac{(a-1)^2}{2!}-b^2+(c-1)\frac{1}{2}\right)+...}{\frac{x^2}{2}-\frac{5}{6}x^3+...}$$

$3 + c - 1 = 0$

$\boxed{c=-2}$ $a - c = 0$

$\Rightarrow a = c = -2$

$$\frac{\frac{9}{2}-b^2-\frac{3}{2}}{\frac{1}{2}}=2$$



$\Rightarrow \frac{9}{2}-\frac{3}{2}-b^2=1$

$\Rightarrow \boxed{b^2=2}$

$a^2 + b^2 + c^2 = 4 + 4 + 2 = 10$

7. If $P\left(10,2\sqrt{15}\right)$ lies on hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and length of latus rectum = 8, then the square of area of ΔPS_1S_2 is [where S_1 & S_2 are the focii of the hyperbola]

(1) 2700 (2) 2400

(3) 1750 (4) 3600

Answer (1)

Sol. $\frac{100}{a^2}-\frac{60}{b^2}=1$

Also $\frac{2b^2}{a}=8$

$\Rightarrow b^2 = 4a$

$\frac{100}{a^2}-\frac{60}{4a}=1$

$100 - 15a = a^2$

$a^2+15a-100=0$

$(a+20)(a-5)=0$

$a = -20, 5$

$\because b^2=4a \Rightarrow a$ cannot be negative

$\Rightarrow \boxed{a=5}$

$\boxed{b^2=20}$

Now $H:\frac{x^2}{25}-\frac{y^2}{20}=1$

$e^2=1+\frac{20}{25}\Rightarrow e=\frac{3}{\sqrt{5}}$

$F:(\pm 3\sqrt{5},0)$ i.e S_1 & S_2

Area $PS_1S_2=\frac{1}{2}\begin{vmatrix}10 & 2\sqrt{15}\\ 3\sqrt{5} & 0\\ -3\sqrt{5} & 0\\ 10 & 2\sqrt{15}\end{vmatrix}=\frac{1}{2}\left[-6\sqrt{75}-6\sqrt{75}\right]$

$=6\sqrt{75}$

$\text{Area}^2 = 2700$

8. If a, b, c are in A.P where $a + b + c = 1$ and a, $2b$, c are in G.P., then the value of $9(a^2 + b^2 + c^2)$ is equal to

(1) 3 (2) –3

(3) 4 (4) –4

Answer (2)

Sol. $a, b, c \rightarrow AP$ Þ

$a, 2b, c \rightarrow$ GP $\Rightarrow 4b^2 = ac$

$a + b + c = 1$

$2b = a + c$

$\Rightarrow b=\frac{1}{3}$

$a+c=\frac{2}{3}$

$4b^2=ac$

$\frac{4}{9}=ac$

$a^2 + b^2 + c^2$

$\Rightarrow \frac{1}{9}+(a+c)^2-2ac$

$\Rightarrow \frac{1}{9}+\frac{4}{9}-2\left(\frac{4}{9}\right)=\frac{-1}{3}$

Here $a=\frac{1}{3}+\frac{i}{\sqrt{3}}\quad b=\frac{1}{3}\quad c=\frac{1}{3}-\frac{i}{\sqrt{3}}$

$\therefore a^2 + b^2 + c^2$ is negative



9. Let the domain of function

$$f(x)=\log_3\log_5(7-\log_2(x^2-10x+15))+\sin^{-1}\left(\left|\frac{3x-7}{17-x}\right|\right)$$

be $(\alpha,\beta]$, then $\alpha+\beta$ is equal to

(1) 6 (2) 7

(3) 8 (4) 9

Answer (4)

Sol. $\log_5(7-\log_2(x^2-10x+85))>0$

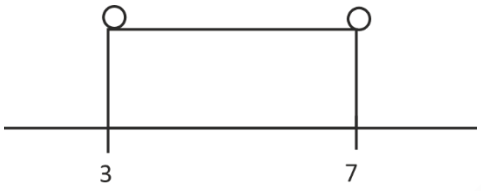
$\Rightarrow 7-\log_2(x^2-10x+85)>1$

$\Rightarrow \log_2(x^2-10x+85)<6$

$\Rightarrow x^2-10x+85<64$

$\Rightarrow x^2-10x+21<0$

$(x-7)(x-3)<0$



$\Rightarrow x \in (3, 7)$

$\left|\frac{3x-7}{17-x}\right|\le 1$

$\Rightarrow |3x-7|\le|17-x|$

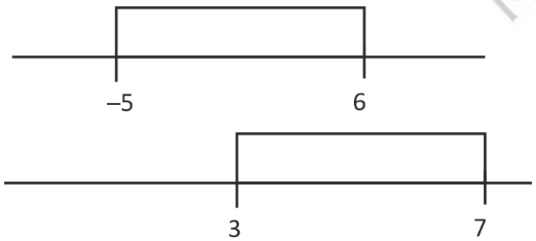
$\Rightarrow 9x^2+49-42x\le 289+x^2-34x$

$\Rightarrow 8x^2-8x-240\le 0$

$\Rightarrow x^2-x-30\le 0$

$(x-6)(x+5)\le 0$

$x\in[-5,\ 6]$



$x \in (3, 6]$

10. Let $\vec{a}=\sqrt{2}\hat{i}$ and $\vec{b}=5\hat{j}+\hat{k}$. If $\vec{c}=\vec{a}\times\vec{b}$ and $\vec{d}$ which lie on *y-z* plane such that $|\vec{d}|=2$. Then the maximum value of $|\vec{c}\cdot\vec{d}|^2$ is equal to

(1) 104 (2) 52

(3) 208 (4) 120

Answer (3)

Sol. $|\vec{a}\times\vec{b}|=\sqrt{52}=|\vec{c}|$

The maximum value of $|\vec{c}\cdot\vec{d}|^2$

$=\left(\sqrt{52}\cdot(2)\cos\theta\right)^2=208\cos^2\theta$

The maximum value of $|\vec{c}\cdot\vec{d}|^2=208$

11. If $\int_0^{64}\left(x^{\frac{1}{3}}+\text{floor}\left(x^{1/3}\right)\right)dx=\alpha$ and $\int_0^{\frac{\pi}{2}}\frac{\sin^2 x}{\sin^6 x+\cos^6 x}dx=\pi\beta$, then the value of ab^2 is equal to

(1) 87 (2) 77

(3) 67 (4) 57

Answer (1)

Sol. $I_1=\int_0^{64}\left(x^{\frac{1}{3}}+\text{floor}\left(x^{\frac{1}{3}}\right)\right)dx$

$=\left.\frac{x^{\frac{4}{3}}}{\frac{4}{3}}\right|_0^{64}+\int_0^1 0\,dx+\int_1^8(1)\,dx+\int_8^{27}2\,dx+\int_{27}^{64}(3)\,dx$

$= 192 + 0 + (8 – 1) + 2) + 2(27 – 8) + 3(64 – 27) = 348$

$I_2=\int_0^{\frac{\pi}{2}}\frac{\sin^2 x}{\sin^6 x+\cos^6 x}dx$



$$\Rightarrow I_2 = \int_0^{\frac{\pi}{2}} \frac{\cos^2 x}{\sin^6 x + \cos^6 x} dx$$

$$\Rightarrow 2I_2 = \int_0^{\frac{\pi}{2}} \frac{1}{\sin^6 x + \cos^6 x} dx = \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{1 + \tan^6 x} dx$$

$$\Rightarrow I_2 = \frac{\pi}{2} \Rightarrow \beta = \frac{1}{2},\ \alpha = 348$$

$$\alpha\beta^2 = 348 \times \frac{1}{4} = 87$$

12.

13.

14.

15.

16.

17.

18.

19.

20.

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. If α, β are roots of equation $12x^2 - 20x + 3\lambda = 0$, $\lambda \in I$. If $\frac{1}{2} \le |\beta - \alpha| \le \frac{3}{2}$, then the sum of all possible values of λ is

Answer (03.00)

Sol. $|\beta - \alpha|^2 \in \left[\frac{1}{4}, \frac{9}{4}\right]$

$$(\alpha^2 + \beta^2 - 2\alpha\beta) = (\alpha + \beta)^2 - 4\alpha\beta \in \left[\frac{1}{4}, \frac{9}{4}\right]$$

$$= \left(\frac{20}{12}\right)^2 - 4\left(\frac{3\lambda}{12}\right) \in \left[\frac{1}{4}, \frac{9}{4}\right]$$

$$\Rightarrow \frac{25}{9} - \lambda \in \left[\frac{1}{4}, \frac{9}{4}\right]$$

$$\Rightarrow 100 - 36\lambda \in [9, 81]$$

$$36\lambda - 100 \in [-81, -9]$$

$$36\lambda \in [19, 91]$$

$$\lambda \in \left[\frac{19}{36}, \frac{91}{36}\right]$$

then integral values of λ is 1, 2

$\Rightarrow$ Sum of integral values of λ is 1 + 2 = 3

22. If $\cos(\alpha + \beta) = -\frac{1}{10}$ and $\sin(\alpha - \beta) = \frac{3}{8}$, where $0 < \alpha < \frac{\pi}{3}$ and $0 < \beta < \frac{\pi}{4}$, if $\tan 2\alpha = \frac{3\left(1 - r\sqrt{55}\right)}{\sqrt{11}\left(s + \sqrt{5}\right)}$ and $r, s \in N$, then $r^2 + s$ is

Answer (20)

Sol. $0 < \alpha < \beta < \frac{7\pi}{12}$

$\because \cos(\alpha + \beta) = -\frac{1}{10}$ $\Rightarrow \alpha + \beta \in$ II[nd] quadrant

$$\sin(\alpha + \beta) = \frac{\sqrt{99}}{10}$$

$$\sin(\alpha - \beta) = \frac{3}{8} \qquad \alpha + \beta \in \text{I quadrant}$$

$$\cos(\alpha - \beta) = \frac{\sqrt{55}}{8}$$

$$\tan\left((\alpha + \beta) + (\alpha - \beta)\right) = \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta)\tan(\alpha - \beta)}$$



$$\tan 2\alpha = \frac{-\sqrt{99}+\frac{3}{\sqrt{55}}}{1+3\sqrt{\frac{99}{55}}} = \left[\frac{-3\sqrt{11.55}+3}{\sqrt{55}+9\sqrt{11}}\right]$$

$$= \frac{3-3\sqrt{11}\cdot\sqrt{55}}{\sqrt{55}+9\sqrt{11}} = \frac{3(1-\sqrt{11}\sqrt{55})}{\sqrt{11}(\sqrt{5}+9)}$$

$r^2 = 11$

$s = 9$

$r^2 + s = 20$

23.

24.

25.

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